D 110234 (Pages : 2) Name.....

Reg. No.....

FIFTH SEMESTER (CBCSS—UG) DEGREE EXAMINATION NOVEMBER 2024

Physics/Applied Physics

PHY 5B 06/APH 5B 06—COMPUTATIONAL PHYSICS

(2019 Admission onwards)

Time : Two Hours Maximum : 60 Marks

The symbols used in this question paper have their usual meanings.

Section A (Short Answer Type)

Answer all questions in two **or** *three sentences.*
Each correct answer carries a maximum 2 *marks.*

1. What is an algorithm ? Write an algorithm to read two numbers and print their sum.
2. What are the different data types in python ?
3. How to add comments in python. What is the importance of comments in programming ?
4. What is a tuple' ? How is it different from 'list'.
5. Write down Newton-Raphson formula.
6. What is the significance of computers in numerical simulations ?
7. Write down the output of the following :
   (a) 3.4//3.
   (b) 3.4 % 3.
8. How do you create a two dimensional array using Numpy ? Give an example.
9. Write the output of the following commands
   (a) print("hello", "world", sep="---").
   (b) x=2; x+=2; print(x).
10. Write a program, using a function to read a velocity in kilometers per hour and print it in meters per second.

Turn over

11. Write a program to create a one dimensional array of numbers from 0 to 9 using numpy.

12. Write down the commands for labeling axes and choosing line styles in plots using Matplotlib.

(Ceiling - 20)

Section B (Paragraph / Problem Type)

(*Answer* **all** *questions in a paragraph of about* **half a page** *to* **one page**, *each correct answer carries a maximum of* 5 *marks*).

13. Write a program to draw the position time graph for a freely falling object.

14. Find the real root of the equation $x^3 - x - 11 = 0$ by using the bisection method.

15. Apply Runge Kutta method to find an approximate value of $y\ (0.\ 2) = 0$, given $y' = x + y$ and $y\ (0) = 1$.

16. Write a python program to read two 3 × 3 matrices and print the matrix product of them. Write a program using Numpy for the same.

17. Write a program to solve $y' = \cos(x)$, given $y\ (0) = 0$. Add code to plot the solution.

18. Evaluate $I = \int_0^6 \frac{1}{1+x}\,dx$ using Simpson's rule.

19. Write a program to graphically simulate the radioactive decay of an element where initial quantity and half life is given.

(Ceiling - 30)

Section C (Essay Type)

Essays - Answer in about **two pages**, *any* one *question.*
The question carries 10 *marks.*

20. Explain Newton's interpolation formula. The population of a town in the census is as given below. Estimate the population for the year 1965 using Newton's forward interpolation formula.

Year	:	1961	1971	1981	1991	2001
Population	:	46	66	81	93	101

21. Given $\frac{dy}{dx} = x^2 - y,\ y(0) = 1,$ find $y(0.1)$ correct to 3 decimal places using Euler method and modified euler method. Write programs for both the methods.

(1 × 10 = 10 marks)